

School of Science and Engineering

AI 501 Mathematics for Artificial Intelligence

ASSIGNMENT 2

Due Date: 1 pm, Friday, October 31, 2025.

Format: 5 problems for a total of 100 marks

Instructions:

- You are allowed to collaborate with your peers but copying your colleague's solution is strictly prohibited. This is not a group assignment. Each student must submit his/her own assignment.
- Solve the assignment on blank A4 sheets and staple them before submitting.
- Submit in-class or in the dropbox labeled AI-501 outside the instructor's office.
- **Write your name and roll no. on the first page.**
- Feel free to contact the instructor or the teaching assistants if you have any concerns.

- You represent the most competent individuals in the country, do not let plagiarism come in between your learning. In case any instance of plagiarism is detected, the disciplinary case will be dealt with according to the university's rules and regulations.
- We require you to acknowledge any use or contributions from generative AI tools. Include the following statement to acknowledge the use of AI where applicable.

I have used [insert Tool Name] to [write, generate, plot or compute; explain specific use of generative AI] [number of times].

Problem 1 (20 marks)**Polynomial Regression and the Bias-Variance Trade-off**

Consider a dataset with a single feature x and a target variable y . We model the relationship using polynomial regression of degree M .

- (a) For a given data point x_i , the prediction is

$$\hat{y}_i = \theta_0 + \theta_1 x_i + \theta_2 x_i^2 + \cdots + \theta_M x_i^M.$$

Given n data points, write the vectorized form of the model $y = A\theta$. Define the design matrix A and explain how its structure and condition number are influenced by the polynomial degree M .

- (b) The regularized cost function for Ridge regression is given by:

$$J(\theta) = \frac{1}{2n} \|y - A\theta\|_2^2 + \frac{\lambda}{2} \|\theta\|_2^2.$$

Derive the closed-form solution for the parameter vector θ that minimizes $J(\theta)$.

- (c) Explain the impact of a very large value of the regularization parameter λ on the bias and variance of the polynomial regression model.
- (d) How does the choice of M interact with λ in this context?

Problem 2 (20 marks)**Comparing Ridge and Lasso Regularization**

You are fitting a linear model $y = Xw + \varepsilon$ with

$$X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}, \quad y = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}.$$

- (a) Write the cost functions for both Ridge and Lasso regression.
- (b) Compute the Ridge regression weight vector w analytically for $\lambda = 1$.
- (c) Explain qualitatively how the Lasso solution differs from Ridge in terms of sparsity.
- (d) Plot or describe the geometric intuition of L1 vs L2 constraints (using contour and constraint shapes).
- (e) Discuss how the choice of λ affects model interpretability and generalization.

Problem 3 (20 marks)**Eigenvalue Decomposition (EVD)**

Consider the following covariance matrix of a 2-dimensional dataset:

$$C = \begin{bmatrix} 4 & 2 \\ 2 & 3 \end{bmatrix}.$$

- (a) Compute the eigenvalues and normalized eigenvectors of C .
- (b) Verify numerically that $C = Q\Lambda Q^T$, where Q is the eigenvector matrix and Λ is the diagonal eigenvalue matrix.
- (c) Comment on the rank of C and the number of independent eigenvectors.
- (d) Explain geometrically what it means for the eigenvectors of C to form a “change of basis” for the linear transformation defined by C .
- (e) Describe the effect of multiplying a vector by C in terms of scaling along eigenvector directions.
- (f) Now, consider the case where if we plot a unit circle in $\mathbb{R}^2$ and apply the transformation defined by C , what would happen to the circle? Explain how the eigenvectors and eigenvalues determine the orientation of the transformed shape (*hint: ellipse*), and the length of its principal axes.

Problem 4 (20 marks)**Singular Value Decomposition (SVD)**

Given a zero-mean data matrix

$$X = \begin{bmatrix} 2 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix}$$

- (a) Compute the SVD of $X = U\Sigma V^T$ explicitly.
- (b) Verify that U and V are orthonormal.
- (c) Construct a rank-1 approximation $X_1 = U_1\Sigma_1V_1^T$ and compute the reconstruction error $\|X - X_1\|_F$.
- (d) Interpret geometrically what the rank-1 approximation represents in the data space.

Problem 5 (20 marks)

Principal Component Analysis (PCA)

Given the following centered dataset:

$$X = \begin{bmatrix} 2 & 0 \\ 0 & 2 \\ 1 & 1 \\ -1 & -1 \\ -2 & 0 \end{bmatrix}$$

- (a) Compute the covariance matrix of X .
- (b) Find the principal components and eigenvalues.
- (c) Project X onto the first principal component (1D).
- (d) Compute how much of the total variance is retained by the first principal component.
- (e) Present a visualization of given X as a projection on the first principal component.

— End of Assignment —